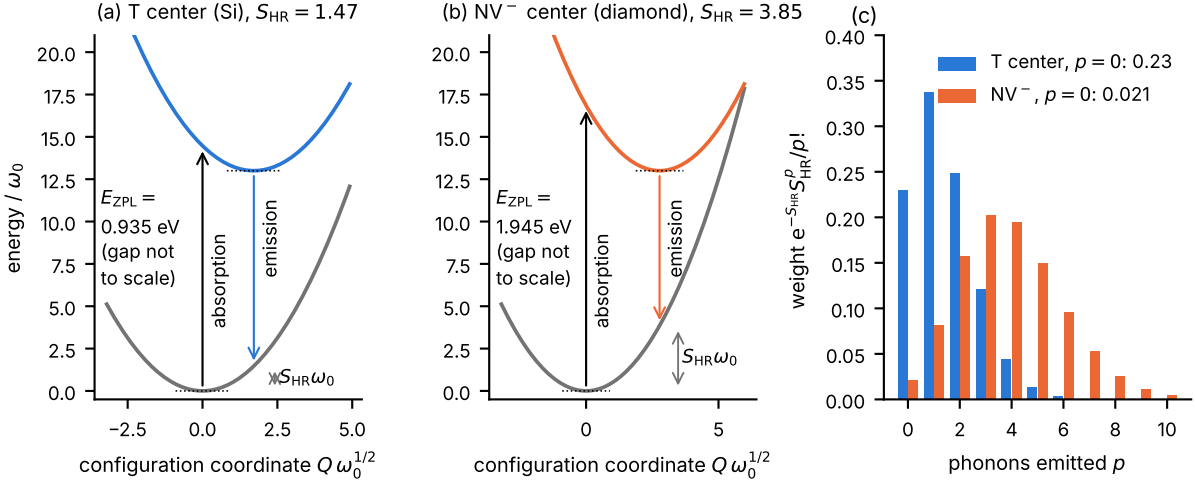
(a) T center (Si), $S_{HR} = 1.47$
(b) NV$^-$ center (diamond), $S_{HR} = 3.85$
(c)
energy / ω_0
E_{ZPL} = 0.935 eV (gap not to scale)
absorption
emission
$S_{HR}\omega_0$
E_{ZPL} = 1.945 eV (gap not to scale)
configuration coordinate $Q\,\omega_0^{1/2}$
weight $e^{-S_{HR}} S_{HR}^p / p!$
T center, $p = 0$: 0.23
NV$^-$, $p = 0$: 0.021
phonons emitted p